

Pavan Kumar Valluri

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DATA ENGINEER

PROFESSIONAL SUMMARY

- **7+ Years of Experience as a Data Engineer** with strong experience in utilizing big data tools for building and automating large-scale data pipelines and application development using Java.
- Strong programming experience developing **REST services**, automation scripts, and data engineering jobs using **Java, Scala, and Python**.
- Experience in developing ETL program for supporting Data Extraction, transformations and loading using Informatica Power Center.
- Experienced in debugging and Performance tuning of the Informatica mappings, sessions and workflows.
- Good understanding of the architecture of Distributed Systems and Parallel processing frameworks for scalable data storage and distributed data processing.
- Utilized various Hadoop ecosystem tools like **HDFS, MapReduce, Hive, Yarn, Kafka, Oozie, Sqoop, and HBase**.
- Strong experience using the Spark framework for performing various data cleansing, enrichment, and aggregation activities.
- Strong experience using Scala and Python to build end-to-end Spark applications.
- Strong experience in troubleshooting Spark application failures and fine-tuning long-running jobs.
- Utilized Spark RDD APIs, **Spark Data Frames, Spark SQL, and Spark Streaming APIs**.
- Strong experience using broadcast variables, accumulators, partitioning, reading text files, **JSON files, and Parquet files**, and fine-tuning various configurations in Spark.
- Hands-on experience writing **Kafka producers** and consuming real-time streaming messages from Kafka using Spark Streaming.
- Worked with significant distributions like **Cloudera & Hortonworks**, including administering the clusters using Cloudera Manager and **Apache Ambari**, respectively.
- Strong experience working with **AWS Cloud services** for building **data lakes** and automating **data pipelines**.
- Utilized AWS Cloud services like **S3, EMR, Glue Metastore, Athena, DynamoDB, and Redshift** extensively.

- Strong experience automating the launching of EMR Clusters and terminating EMR Clusters using Terraform and AWS Java SDK.
- Experience working on creating and running Docker Images using ECS for orchestration.
- Strong experience using Sqoop for performing bulk imports and incremental loads of relational data from various databases.
- Strong experience using **Oozie** and **AWS Step Functions** to orchestrate and schedule various data ingestion and processing jobs.
- Experienced in writing custom Hive UDFs to incorporate business logic with Hive Queries.
- Strong experience fine-tuning long-running Hive queries utilizing Partitioning, Bucketing, Columnar file formats, and other configuration optimizations.
- Strong experience using **NoSQL Databases like HBase and DynamoDB**.
- Experience in JIRA for tracking test results and interacting with developers to resolve issues.
- Strong Java developer and developed REST applications using Spring Boot.
- Detailed understanding of **Software Development Life Cycle (SDLC)** and sound knowledge of project implementation methodologies, including Scrum, Waterfall, and Agile.
- Active team player with excellent interpersonal skills and a keen learner with self-commitment & innovation.
- Superior communication skills, strong decision-making and organizational skills, and outstanding analytical and problem-solving skills to undertake challenging jobs.
- Certified in **AWS Solutions Architect – Associate** and **Azure Data Engineer Associate**.
- Expertise in real-time data processing using **Kafka** and **Spark Streaming**.
- Strong experience in data migration from on-premises systems to cloud platforms like **AWS** and **Azure**.
- Proficient in data security practices, including PII encryption and role-based access control (RBAC).
- Extensive experience in **ETL tools like Informatica PowerCenter** for designing, developing, and deploying ETL workflows.
- Strong experience in **Informatica Data Integration** for extracting, transforming, and loading data from heterogeneous sources.
- Proficient in creating **Informatica mappings, workflows, and sessions** to process large volumes of data efficiently.
- Strong experience in **ETL pipeline development** using Informatica for data warehousing and data lake solutions.

- Experience in orchestrating and scheduling data pipelines using **Apache Airflow** for workflow automation and monitoring.
- Proficient in building interactive dashboards and reports using **Tableau** and **Power BI** for data visualization and business insights.

TECHNICAL EXPERTISE

Category	Skills
Big Data Technologies	HDFS, MapReduce, Hive, Pig, Sqoop, Flume, Oozie, Zookeeper, Kafka, MongoDB, Spark, Spark Streaming, HBase, Flume, Impala
Hadoop Distribution	Cloudera, Hortonworks
Cloud Tools	AWS (S3, EMR, Redshift, Glue, Lambda), Azure (Data Factory, Synapse Analytics, Databricks, Data Lake Storage, Blob Storage)
Languages	Java, SQL, Scala, Python, Pig Latin, HiveQL
Databases	Oracle, SQL Server, MySQL, NoSQL, Snowflake, Teradata
DevOps Tools	Docker, Kubernetes, Jenkins, Terraform, Git, CI/CD pipelines
File Formats	Parquet, Avro, ORC, JSON, CSV
BI Tools	Power BI, Tableau
ETL Tools	Informatica PowerCenter, Informatica Cloud (IICS), Informatica Data Quality (IDQ)

PROFESSIONAL EXPERIENCE

Client: Express Scripts, NJ

Title: Sr. Azure Data Engineer

Duration: January 2024 – Present

- Involved in writing **PySpark** applications to perform various data cleansing, validation, transformation, and summarization activities according to requirements.
- Responsible for developing data pipelines using **Azure Data Factory** involving investigating raw JSON files, transactional and user profile information from on-prem data warehouses and processing them using PySpark and Scala.
- Written Python, PySpark, and Scala transformations using Jupyter and Databricks on data loaded in **Azure Data Lake Storage**.
- Using **Synapse** Dedicated SQL pool Connector for writing the processed data from Spark to dedicated SQL pool.
- Written Python and Spark transformations on raw data to clean it.
- Wrote Python validation code to check the consistency of data ingesting.
- Worked extensively on improving performance and troubleshooting Spark and Delta Tables data pipelines.
- Implemented Dockerized Spark notebook, which supports Docker containerization.
- Involved in the deployment of Docker Container into Kubernetes clusters.
- Implemented data migration from multiple on-premises servers into the cloud using the Data Factory service in Azure and the Data Migration Assistant application.
- Implemented **Azure Synapse** pipelines to gather data from SQL On-prem and cloud Databases.
- Worked with SSMS & SSIS packages to extract data from on-prem databases.
- Worked on PII data encryption SSIS packages while working from customer-facing databases.
- Worked on data analytics with Power BI to check data quality and correctness.
- Worked on the **Azure Event Grid** to collect data from sensors.
- Implemented **Azure Functions** and Logic Apps to run serverless apps.
- Worked with Hive and SQL Server to store and capture data using data pipelines.
- Written Spark transformations in **Synapse** notebooks to clean the raw data.
- Worked on different compressed file formats like Parquet, AVRO, and ORC using Spark.
- Snowflake was a data warehousing system that stored processed data from Databricks and **ADF**.
- Migrated **10+ TB** of on-premises data to **Azure Synapse Analytics**, reducing query response times by **30%**.

- Optimized **Spark jobs**, reducing job execution time by **25%**.
- Built **Power BI dashboards** for data visualization and reporting.
- Analyzed the business requirements and framing the Business Logic for the ETL Process and maintained the ETL process using InformaticaPowerCenter 10.4.0
- Designed and developed **ETL workflows** using **Informatica PowerCenter** to extract, transform, and load data from multiple sources into Azure Synapse.
- Utilized **Informatica Cloud (IICS)** for cloud-based data integration and ETL processes, ensuring seamless data flow between on-prem and cloud systems.
- Developed interactive dashboards and reports using **Tableau** and **Power BI** for business insights and decision-making.
- Orchestrated and scheduled data pipelines using **Apache Airflow** for workflow automation and monitoring.

Environment: Azure cloud, Data Factory, Data Lake Storage Gen2, Synapse, Spark, SQL, GitHub, GitHub CI/CD, Snowflake, Docker, Kubernetes, Informatica PowerCenter, Informatica Cloud (IICS), Apache Airflow, Tableau, Power BI.

Client: FINRA

Title: Sr. Data Engineer

Duration: May 2023 – December 2023

- Involved in requirement gathering, business analysis, design and development, testing, and implementation of business rules.
- Understand business use cases, integration business, write business & technical requirements documents, logic diagrams, process flow charts, and other application-related documents.
- Used Pandas in Python for Data Cleansing and validating the source data.
- Created pipelines, data flows, complex transformations, and manipulations using Glue and PySpark.
- Developed ETL applications using Python, Spark (PySpark), and Shell scripting based on business requirements.
- Involved with Data Profiling for multiple sources and answered complex business questions by providing data to business users.
- Wrote complex SQL queries to validate the data against different kinds of database systems and reconcile data across systems such as Snowflake and Oracle.
- Developed CI/CD system with Jenkins on Docker for the runtime environment for the CI/CD system to build, test, and deploy.
- Worked on PySpark jobs and troubleshoot PySpark job performance. Worked on Batch Data pipelines using PySpark and Scala.

- Good experience working with Databricks Notebook platform and Snowflake Cloud Data Warehouse systems.
- Worked on building a centralized Data Lake on AWS Cloud by utilizing services like **SQS, SNS, S3, EMR, RedShift, and Athena**.
- Leveraged Snowflake's Snow Sight and Snow SQL to ingest and build data models for downstream analytics.
- Worked on Data Ingestion from external systems into an S3 data lake using **Python** and **boto3** module.
- Worked on Docker files to containerize the Spark application.
- Automated **CI/CD pipelines**, reducing deployment time by **50%**.
- Optimized **Redshift queries**, improving query performance by **25%**.
- Collaborated with data scientists to build machine learning pipelines using **SageMaker**.
- Designed and implemented **ETL workflows** using **Informatica PowerCenter** to integrate data from multiple sources into Snowflake.
- Utilized **Informatica Data Quality (IDQ)** for data profiling and cleansing, ensuring data accuracy and consistency.

Environment: AWS EMR, ECS, Lambda, Spark, Snowflake, S3, Athena, Kafka, HBase, Scala, Redshift, Step Functions, Docker, Informatica PowerCenter, Informatica Data Quality (IDQ).

Client: West Agile Labs, India

Title: Hadoop/Spark Developer

Duration: July 2020 – November 2022

- Developed and maintained a **Data Lake** containing regulatory data for federal reporting using big data technologies such as Hadoop Distributed File System (**HDFS**), **Apache Impala, Apache Hive, and Cloudera distribution**.
- Developed different **ETL** jobs to extract data from various sources like **Oracle** and **Microsoft SQL Server**, transform the extracted data using Hive Query Language (**HQL**), and load it into the Hadoop Distributed file system (**HDFS**).
- Involved in importing the data from different sources into **HDFS** using **Sqoop** and applying transformations using **Hive and Spark**, then loading data into Hive tables.
- Fixed data-related issues within the Data Lake.
- Primarily involved in Data Migration process using **AWS** by integrating with GitHub repository and Jenkins.
- Designed, developed, and maintained data integration programs in a Hadoop and RDBMS environment with traditional and non-traditional source systems and **RDBMS** and **NoSQL** data stores for data access and analysis.

- Performed Spark's in-memory computing capacity to perform procedures such as text analysis and processing using **Scala**.
- Primarily responsible for designing, implementing, Testing, and maintaining database solutions for **AWS**.
- Experience working with **Spark Streaming** and dividing data into different branches for batch processing through the **Spark engine**.
- Implementing new functionality in the Data Lake using big data technologies such as Hadoop Distributed File System (HDFS), Apache Impala, and Apache Hive based on the requirements provided by the client.
- Communicating regularly with the business teams and the project manager to resolve gaps between the client's and the project's technical requirements.
- Developing **Python scripts** using Hadoop Distributed File System APIs to generate **Curl commands** to migrate data and prepare different project environments.
- Coordinating the Production releases with the change management team using the Remedy tool.
- Communicating effectively with team members and conducting code reviews.
- Migrated on-premises workloads to **AWS**, reducing infrastructure costs by **20%**.
- Optimized **Spark jobs**, reducing job execution time by **30%**.
- Automated data workflows using **Oozie** and **AWS Step Functions**.

Environment: Hadoop, Data Lake, AWS, Python, Spark, Hive, Cassandra, ETL Informatica, Cloudera, Oracle 10g, Microsoft SQL Server, Control-M, Linux.

Client: Novartis, India

Title: Java Developer

Duration: January 2017 – June 2020

- Involved in Requirements Analysis and design of an Object-oriented domain model.
- Involvement in the detailed Documentation and written functional specifications of the module.
- Involved in the development of Applications with Java and J2EE technologies.
- Developed and maintained elaborate services-based architecture utilizing open-source technologies like Hibernate, ORM, and Spring Framework.
- Developed server-side services using Java multithreading, Struts MVC, Java, EJB, Spring, and Webservices (SOAP, WSDL, AXIS).
- Responsible for developing the DAO layer using Spring MVC, configuring XML for Hibernate, and managing CRUD operations (insert, update, and delete).

- Designing, Development, and Implementation of JSPs in the Presentation layer for Submission, Application, and Reference implementation.
- Development of JavaScript for client-end data entry validations and Front-End Validation.
- Deployed Web, presentation, and business components on Apache Tomcat Application Server.
- Developed PL/SQL procedures for different use case scenarios.
- Involvement in post-production support, Testing, and using JUnit for unit testing of the module.
- Optimized database queries, reducing query execution time by **15%**.
- Collaborated with data engineering teams to migrate on-premises applications to **AWS**.

Environment: Java/J2EE, JSP, XML, Spring Framework, Hibernate, Eclipse (IDE), Java Script, Ant, SQL, PL/SQL, Oracle, Windows, UNIX, Soap, Jasper reports.

EDUCATION

Master of Science in Information Technology at University of the Cumberland - Williamsburg, KY